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1. (a) Let $p_1, p_2, \dots, p_n$ be primes. Show that the expression $p_1 p_2 \dots p_n + 1$ is divisible by none of these primes.

Proof. Let $p \in \{p_1, p_2, \dots, p_n\}$. By contradiction, assume

$$p \mid p_1 p_2 \dots p_n + 1.$$

This implies that $\exists s \in \mathbb{Z}$ s.t

$$p_1 p_2 \dots p_n + 1 = ps.$$

But, since p is prime, its divisors can only be 1 or p . Since every prime number is greater than 1, $p \nmid 1$. So, the only way for this expression to be true is if

$$p_1 p_2 \dots p_n + 1 = p$$

which we know cannot be true. $\therefore p \nmid p_1 p_2 \dots p_n + 1$. □

(b) Use part (a) to prove that there are infinitely many primes.

Proof. By contradiction, assume that there are finitely many primes. There must exist some maximum prime, call it p_f . By (a),

$$p_1, p_2, \dots, p_f \nmid p_1 p_2 \dots p_f + 1.$$

Let $m = p_1 p_2 \dots p_f + 1 \in \mathbb{Z}$. By the Fundamental Theorem of Arithmetic, m is either a prime or a product of primes. But if m is a prime,

$$\begin{aligned} p_f &< p_1 p_2 \dots p_f + 1 \\ p_f &< m \end{aligned}$$

contradicts our assumption. So, m must be a product of primes, which implies $\exists$ a prime p_i that divides it. But that contradicts (a) and $\therefore m$ can not be a product of primes. Thus, this contradicts the Fundamental Theorem of Arithmetic, and shows that the assumption of p_f being the largest prime must be false. $\therefore$ there must be infinitely many primes. □

2. For every positive integer n , prove that a set with exactly n elements has exactly 2^n subsets (including the empty set and the entire set).

Proof. We proceed by induction on n .

Base Case: Let $n = 1$. Then, the set $\{1\}$ has the subsets

$$\underbrace{\emptyset, \{1\}}_{2^1}$$

which amount to 2^1 subsets.

Induction Step: Let $n = k$ where k is some integer greater than 1. Assume the set $S_1 = \{1, 2, 3, \dots, k\}$ has the subsets

$$\underbrace{\emptyset, \{1\}, \{2\}, \dots, \{1, 2, 3, \dots, k\}}_{2^k}$$

which amount to 2^k subsets. We must show that for $n = k+1$ the set $S_2 = \{1, 2, 3, \dots, k, k+1\}$ has 2^{k+1} subsets. But, since S_1 is already contained in S_2 , all of S_1 's subsets must also be a subset of S_2 . Thus, the current number of subsets in S_2 is

$$2^k + \underbrace{\{1, 2, 3, \dots, k, k+1\} + \dots}_{\text{remaining amount of subsets}}$$

But, the only difference between the subsets of S_1 and the remaining subsets of S_2 is the element $k+1$. So, the remaining subsets, excluding $k+1$, must each be a subset of S_1 which we know has 2^k subsets. So the total subsets of S_2 must be

$$\begin{aligned} 2^k + 2^k &= 2(2^k) \\ &= 2^{k+1}. \end{aligned}$$

So, a set of n elements has 2^n subsets for $n > 0$. □

3. The Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, 34, In general, the Fibonacci numbers are defined by $f_1 = 1$, $f_2 = 1$, and for $n \geq 3$, $f_n = f_{n-1} + f_{n-2}$. Prove that the n th Fibonacci number f_n satisfies $f_n < 2^n$ for all $n \geq 1$.

Proof. We proceed by induction on n .

Base Case: Let $n = 1$. Then,

$$\begin{aligned} f_1 &= 1 \\ \Rightarrow 1 &< 2^1 = 2 \quad \checkmark \end{aligned}$$

Induction Step: Let $n = k$ where $k \in \mathbb{Z}$ and $k \geq 1$. Assume that $f_k < 2^k$. We must show that for $n = k+1$, $f_{k+1} < 2^{k+1}$. By the definition of the Fibonacci numbers, we know that

$$f_{k+1} = f_k + f_{k-1}.$$

But, we know that $f_k < 2^k$. So,

$$f_{k+1} < 2^k + f_{k-1}.$$

But, $f_{k-1} < f_k$ which implies $f_{k-1} < 2^k$. Thus,

$$\begin{aligned} f_{k+1} &< 2^k + 2^k \\ \Rightarrow f_{k+1} &< 2(2^k) \\ \Rightarrow f_{k+1} &< 2^{k+1}. \end{aligned}$$

$\therefore f_n < 2^n$ holds $\forall n \geq 1$ □

4. Let $a, b \in \mathbb{R}$, define $a \sim b$ if $a - b$ is an integer. Show that $\sim$ is an equivalence relation on $\mathbb{R}$. Describe the equivalence classes produced by this relation.

To show that $\sim$ is an equivalence relation on $\mathbb{R}$, we must show that the properties of one hold.

Reflexivity: Let $a \in \mathbb{R}$. We want to show that $a \sim a$. That is,

$$\begin{aligned} a - a &\in \mathbb{Z} ? \\ \Rightarrow 0 &\in \mathbb{Z} \checkmark \end{aligned}$$

Symmetry: Suppose $a \sim b$. We want to show that $b \sim a$. That is,

$$b - a \in \mathbb{Z} ?$$

Since $a \sim b$, $a - b \in \mathbb{Z} \therefore b - a \in \mathbb{Z}$ but with the signs flipped.

Transitivity: Suppose $a \sim b$ and $b \sim c$. We want to show that $a \sim c$. That is

$$a - c \in \mathbb{Z} ?$$

Since $a \sim b$ and $b \sim c$,

$$\begin{aligned} a - b, b - c &\in \mathbb{Z} \\ \Rightarrow (a - b) + (b - c) &\in \mathbb{Z} \\ \Rightarrow a - c &\in \mathbb{Z} \checkmark \end{aligned}$$

Thus, $\sim$ is an equivalence relation on $\mathbb{R}$. Equivalence classes produced by $\sim$ are of the form

$$\underbrace{[a] = \{x.a \mid x, a \in \mathbb{R}, x \sim a\}}_{\text{the class of all real numbers with } a \text{ following the decimal point}}$$

5. Suppose $f : A \rightarrow B$ and $g : B \rightarrow C$ are two functions.

(a) If f and g are one-to-one, show that the function gf (the composition $g \circ f$) is also one-to-one.

Proof. Suppose f and g are one-to-one. Let $a_1, a_2 \in A$ such that

$$gf(a_1) = gf(a_2).$$

Since g is one-to-one, this equality implies that

$$f(a_1) = f(a_2).$$

But, f is also one-to-one. So,

$$a_1 = a_2$$

implies $g \circ f$ is one-to-one. □

(b) If f and g are onto, show that the function gf is also onto.

Proof. Suppose f and g are onto. Let $a_1 \in A$ and $b_1 \in B$ such that

$$gf(a_1) = g(b_1).$$

Since f is onto, $\exists$ some $a_1 \in A$ s.t $f(a_1) = b_1$. But, g is also onto, which implies that $\exists$ some $c_1 \in C$ s.t $g(b_1) = c_1$. Rewriting the composition,

$$gf(a_1) = c_1$$

shows that $g \circ f$ is onto.

□